



00:01

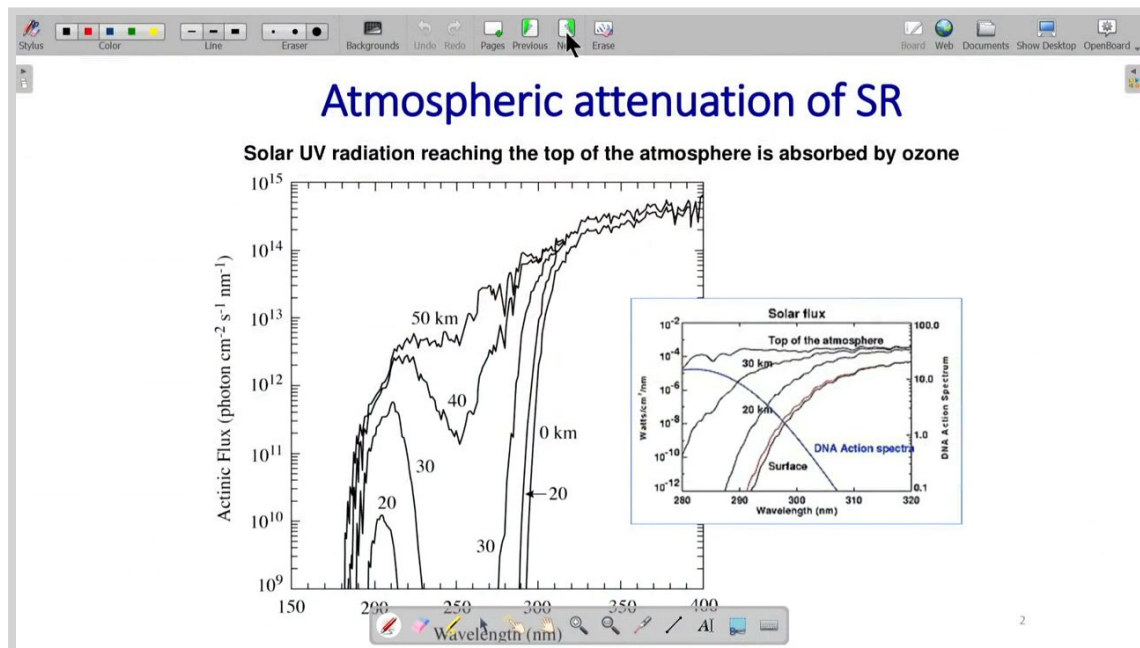
Ozone Layer Depletion

Stratospheric ozone and its importance

– *The 'hole' story*

1

01:02



04:01

Dobson Unit

1 Dobson Unit (DU) is defined to be 0.01 mm thickness at stp; the ozone layer over Labrador is ~300 DU.

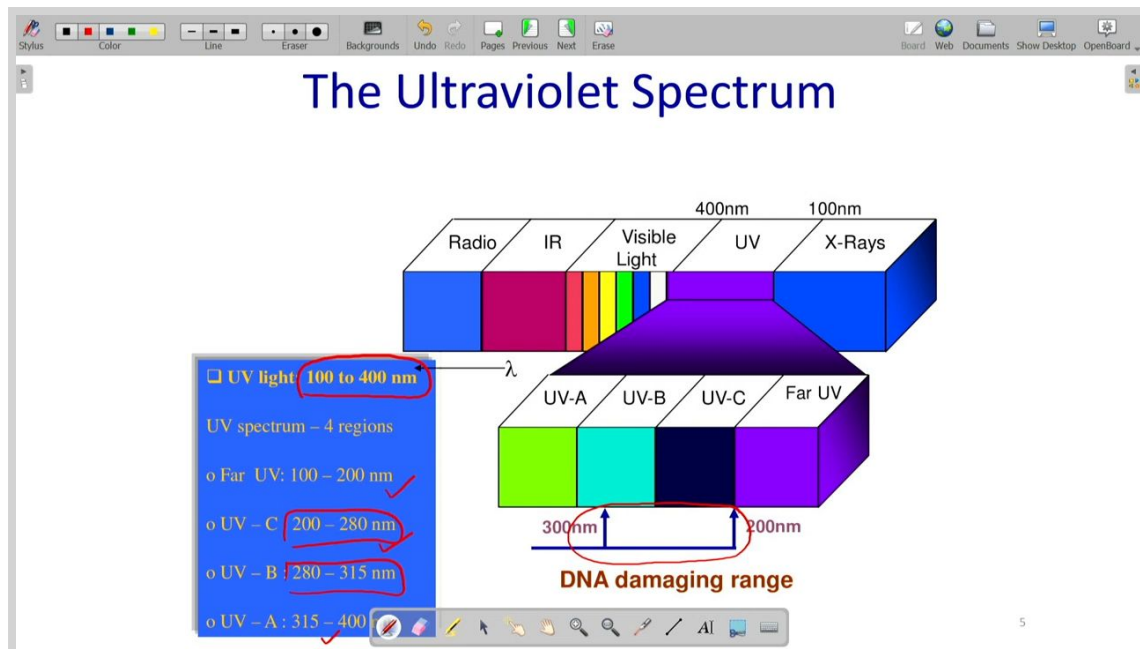
Mean ratio, column O₃: air = 5×10^{-7}

04:39

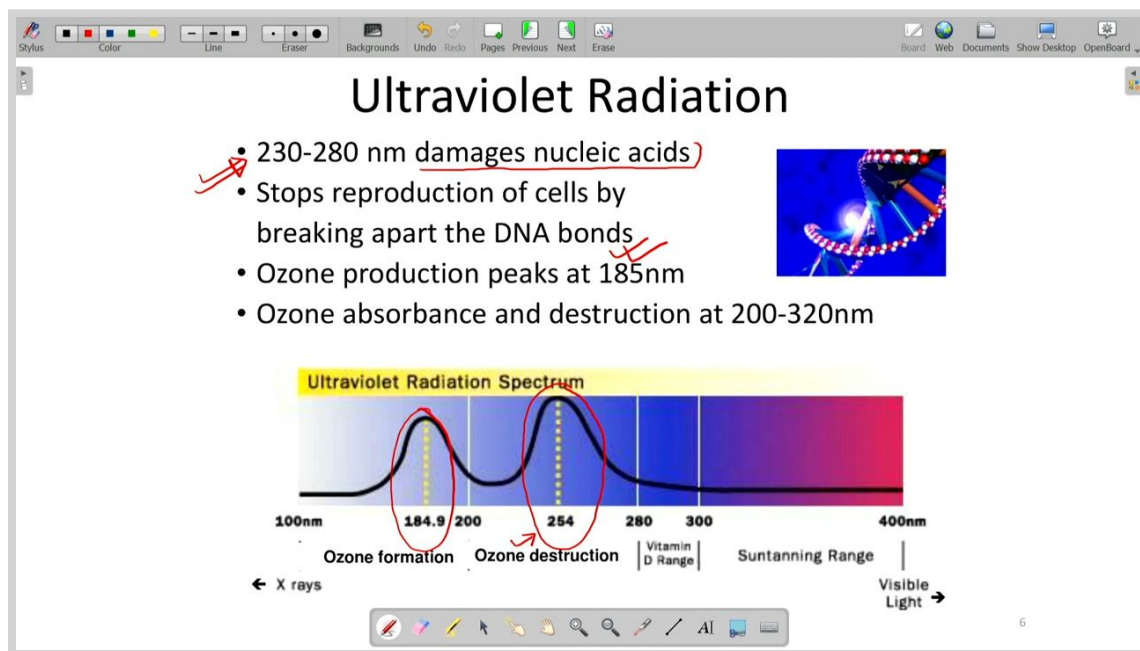
Ozone in Stratosphere: Dobson Units (DU)

If 100 DU of ozone were brought to the Earth's surface, it would form a layer 1 millimeter thick. In the tropics, ozone levels are typically between 250 and 300 DU year-round. In temperate regions, seasonal variations can produce large swings in ozone levels. For instance, measurements in Leningrad have recorded ozone levels as high as 475 DU and as low as 300 DU.

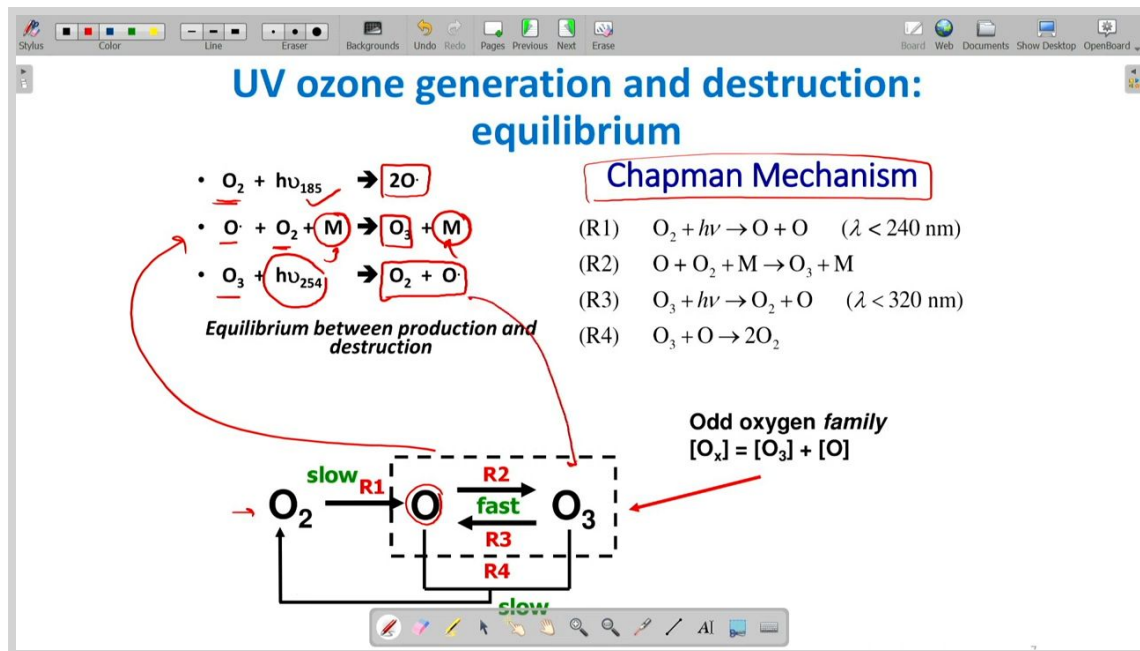
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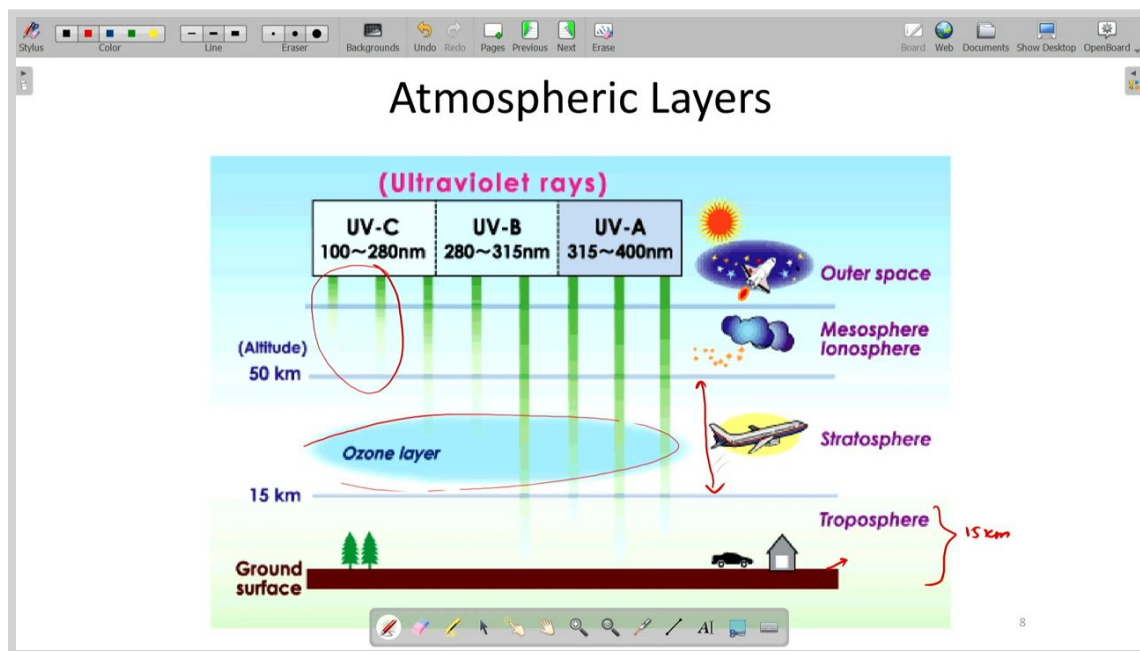
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09:48



13:24

The catalyzed cycle of stratospheric ozone production and destruction

1. $h\nu + \text{O}_2 \rightarrow 2\text{O}\cdot$ 184 nm
2. $\text{O}\cdot + \text{O}_2 \rightarrow \text{O}_3$
3. $\text{O}_3 + \text{X} \rightarrow \text{O}_2 + \text{OX}$
4. $\text{OX} + \text{O}\cdot \rightarrow \text{O}_2 + \text{X}$

X could be Cl from a CFC

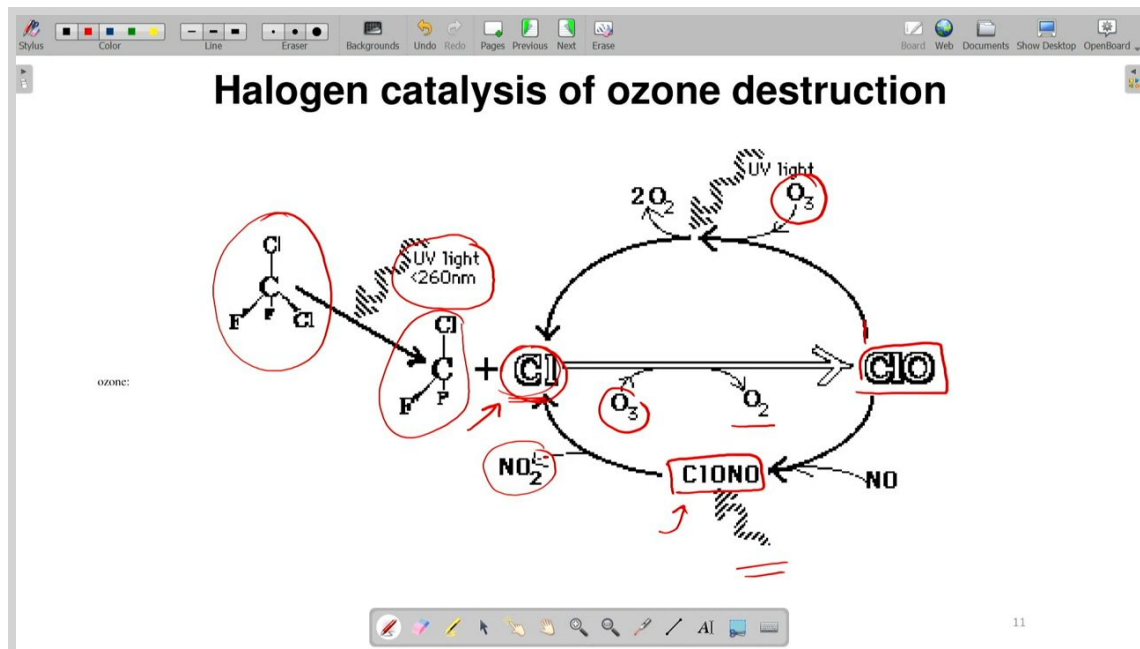
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Ozone Depletion Process

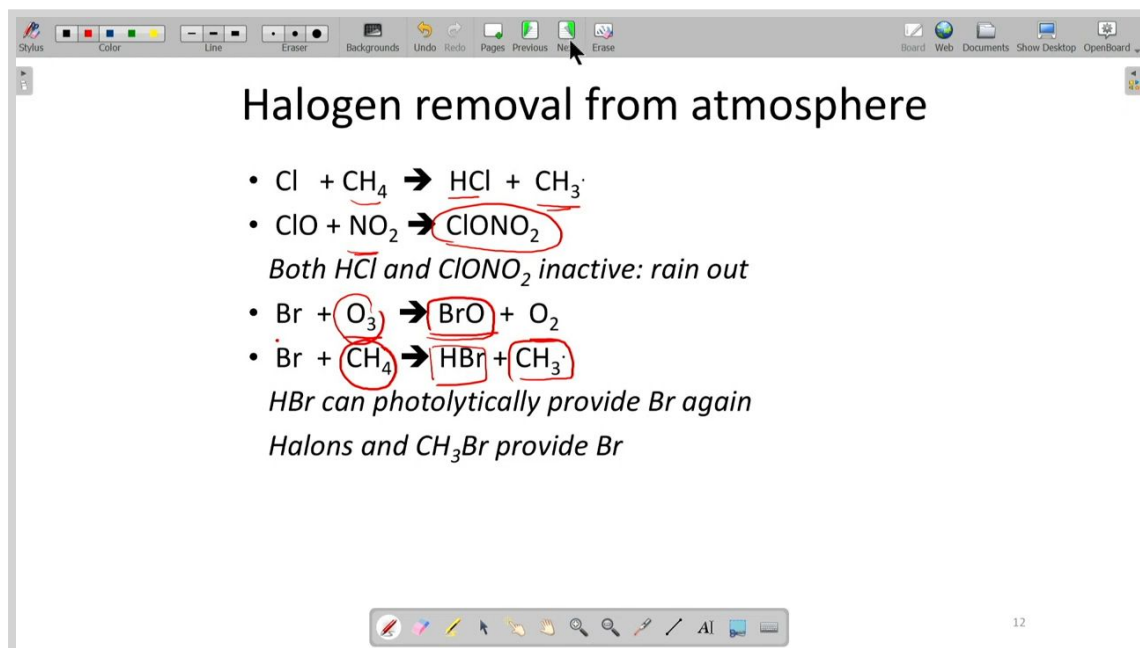
1. CFCs released
2. CFCs rise into ozone layer
3. UV releases Cl from CFCs
4. Cl destroys ozone
5. Depleted ozone → more UV
6. More UV → more skin cancer

Direct destruction of O_3
causing O_3 to stay in the system that causes O_3 formation

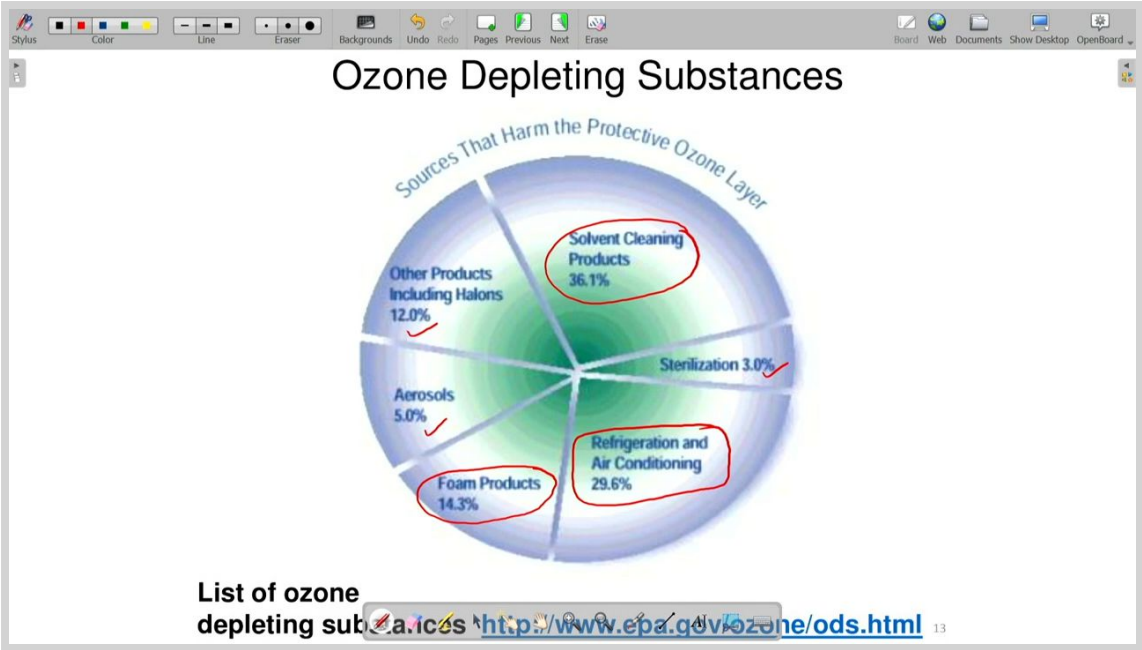
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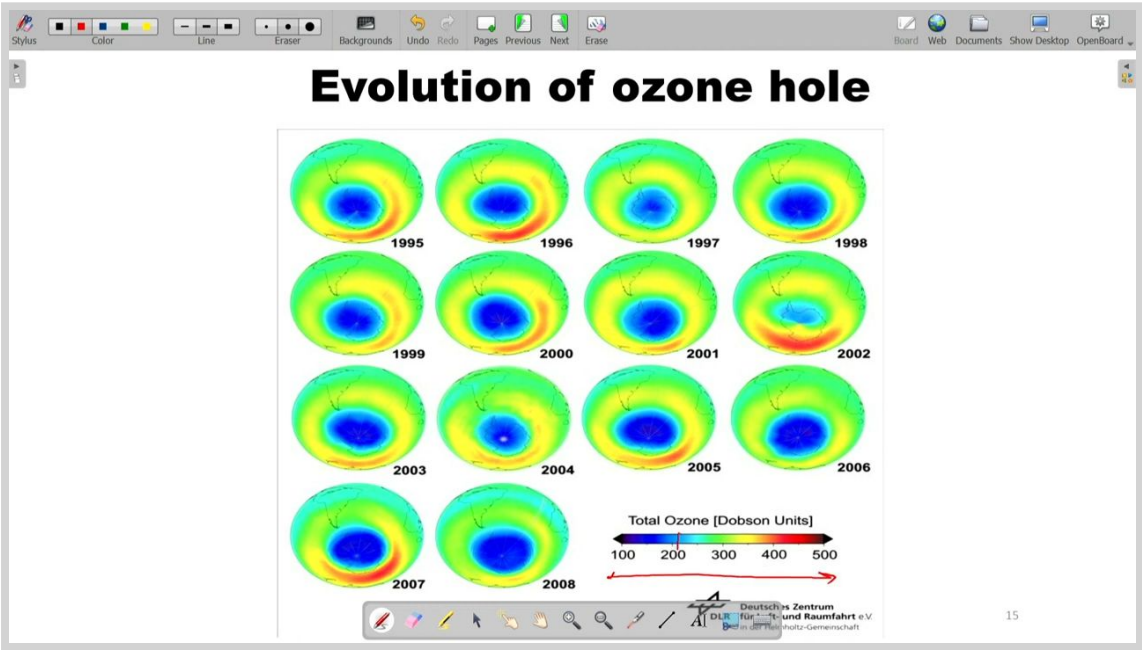
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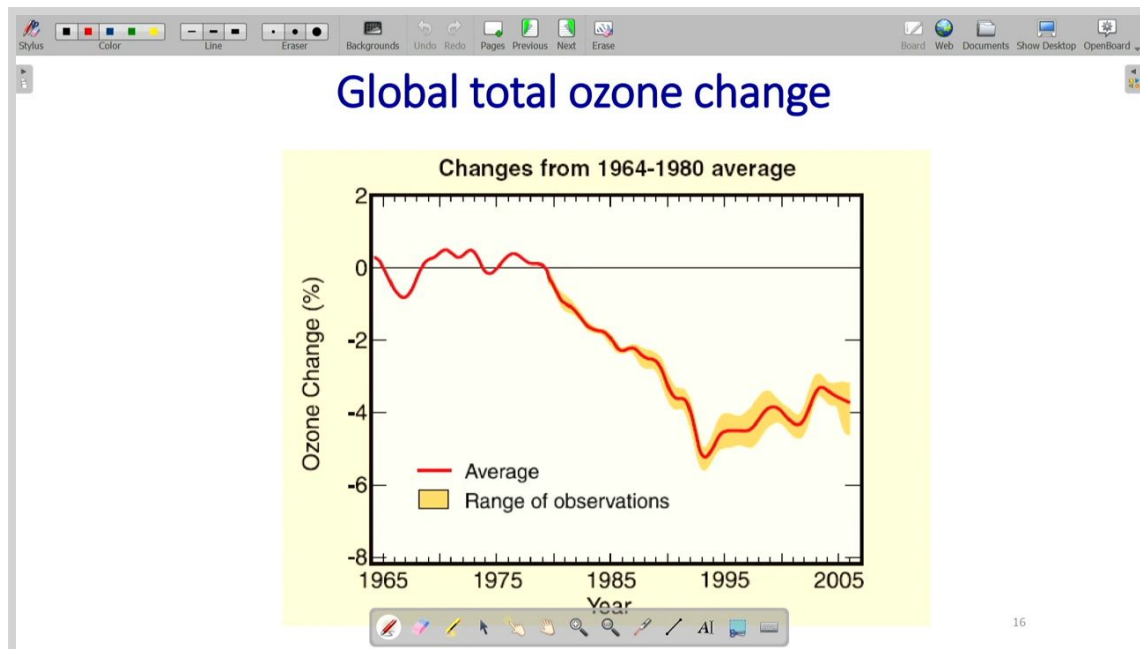
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18:50



20:33



21:04

Noble for Ozone hole explanation

The Nobel Prize in Chemistry 1995




Paul J. Crutzen
The Netherlands
Max-Planck-Institute
for Chemistry Mainz, Germany
1933 -



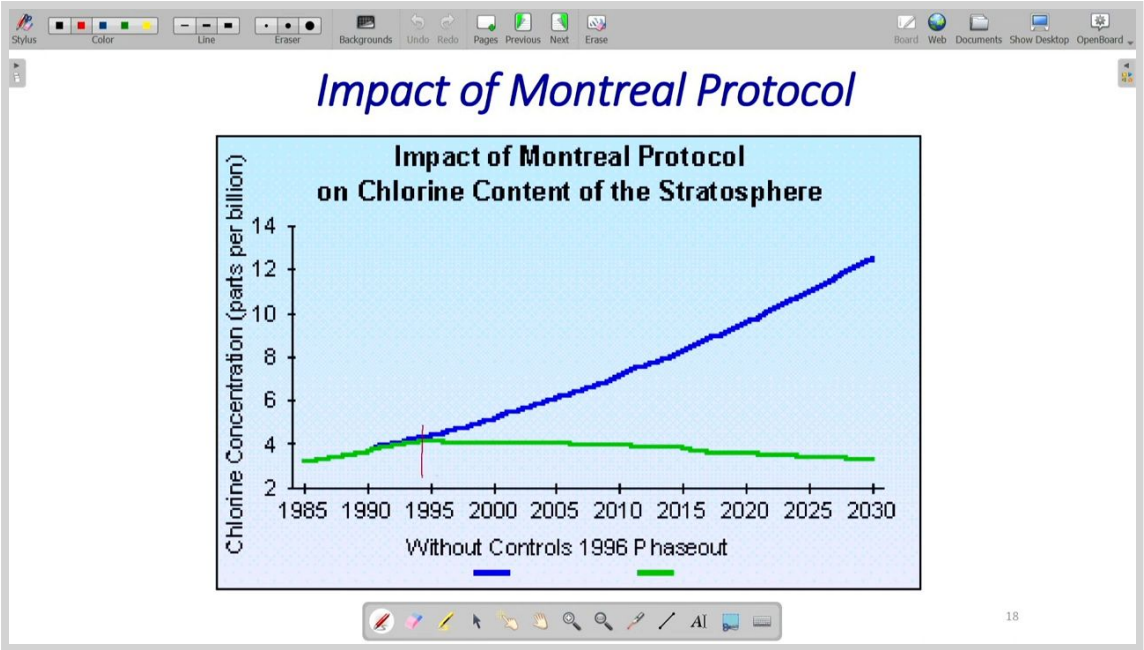
Mario J. Molina
USA
MIT, USA
Cambridge, MA
1943 -



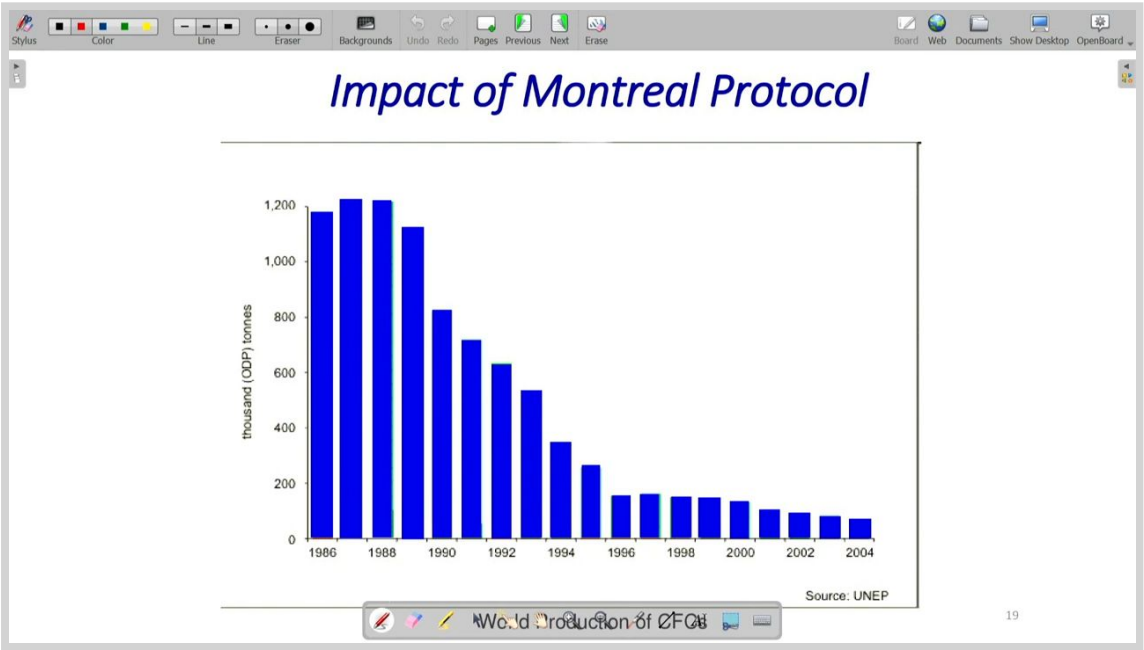
F. Sherwood Rowland
USA
Department of Chemistry,
University of California
Irvine, CA, USA
1927 -

"for their work in atmospheric chemistry,
particularly concerning the formation and
decomposition of ozone"

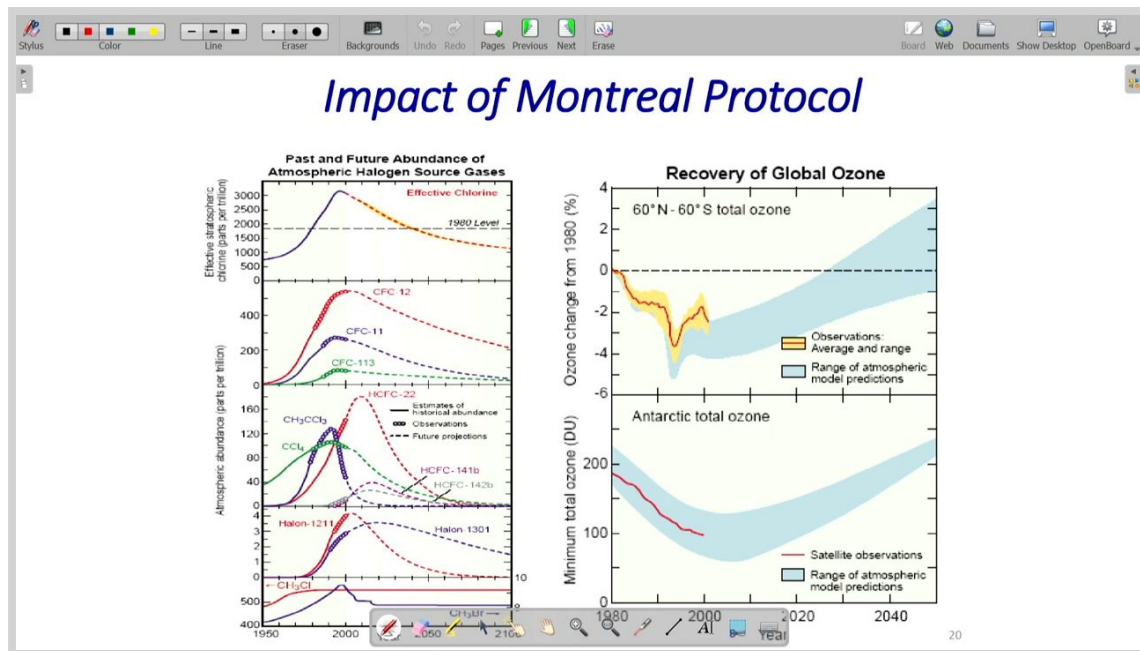
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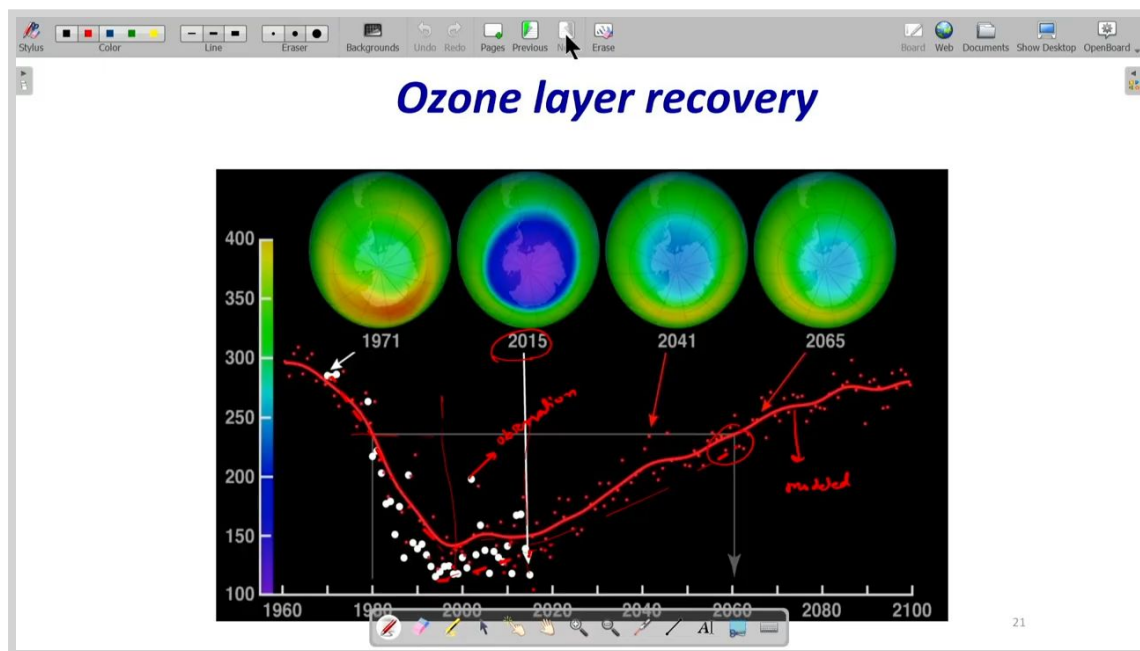
22:09



22:23



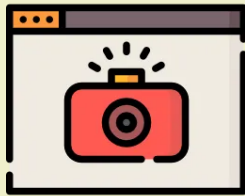
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